

WASP-50b

R-1750

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131027 · created 2026-08-02T19:50:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456592.842 +/- 0.094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	3.01 +/- 1.77 [%]
Orbital Inclination (inc)	87.39 +/- 2.88
Airmass coefficient 1 (a1)	2.75 +/- 0.16
Airmass coefficient 2 (a2)	-0.72 +/- 0.2
Scatter in the residuals of the lightcurve fit is	5.37 %
Best Comparison Star	#2 - [296, 436]
Optimal Aperture	2.34
Optimal Annulus	9.28
Transit Duration (day)	0.048 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.1404 (deviation 0.64σ)
Duration fit vs published	0.048 d vs 0.0754 d (deviation 0.76σ)
Mid-time O–C	-21.3 min
Depth SNR	3.4
Residual scatter	5.37 %

Light curve

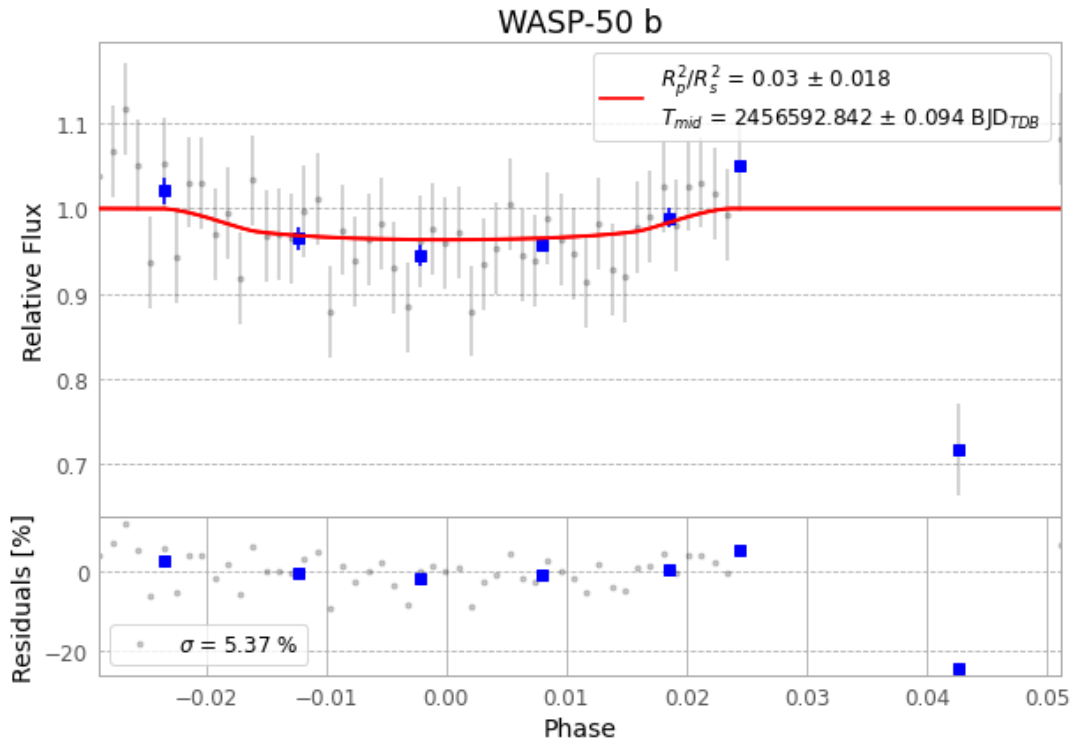


Figure 1 — FinalLightCurve_WASP-50 b_2013-10-26.png (SHA-256 1e5973e9c64865bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b5482ee616be01a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-50131027064212.FITS → WASP-50131027103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131027.log (35a05f0e7a94...), FinalLightCurve_WASP-50 b_2013-10-26.png (1e5973e9c648...), FinalLightCurve_WASP-50 b_2013-10-26.csv (69849bd43756...)

Compute resources

Wall time 385.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 19:50Z.